

## Experiment 8: Keycloak Authentication using Docker

This document contains complete step-by-step instructions to perform Experiment 8 using Docker, Keycloak, Java GUI Client, and Java GUI Server.

### Experiment Objective

To implement secure authentication using Keycloak and JWT tokens for an Online Examination System.

### Software Required

- Docker Desktop
- Java JDK
- VS Code
- Internet Connection
- Keycloak (via Docker)

## PHASE 1: Start Docker and Keycloak

### Step 1: Open Docker Desktop

Start Docker Desktop and wait until "Engine Running" appears.

### Step 2: Open Command Prompt

### Step 3: Run Keycloak Container

Run the following command:

```
docker run -d -p 8080:8080 -e KEYCLOAK_ADMIN=admin -e  
KEYCLOAK_ADMIN_PASSWORD=admin123 quay.io/keycloak/keycloak:latest start-dev
```

### Step 4: Verify Container Running

Run:

```
docker ps
```

You should see Keycloak container running.

### Step 5: Open Keycloak

Open browser:

<http://localhost:8080>

### Step 6: Login to Admin Console

Click "Administration Console"

Username: admin

Password: admin123

## PHASE 2: Configure Keycloak

### Create Realm

Top-left dropdown → Create Realm

Realm Name: OnlineExamRealm

### Create Client

Go to Clients → Create Client

Client ID: exam-app

Client Type: OpenID Connect

Enable Direct Access Grants.

### Create Roles

Go to Realm Roles

Create roles:

- student

- admin

- faculty

### Create User

Go to Users → Add User

Username: student1

### Set Password

Go to Credentials

Password: 1234

Temporary: OFF

### Assign Role

Go to Role Mapping

Assign role: student

## PHASE 3: Java Project Setup

### Create Project Folder

Create folder:

IS\_exp8

### Add Java Files

Add these files inside folder:

- ExamClientGUI.java

- ExamServerGUI.java
- DESUtil.java

### Open VS Code

Open VS Code → Open Folder → IS\_exp8

### Compile Code

Open terminal in VS Code and run:

```
javac *.java
```

## PHASE 4: Run Project

### Run Server

Run:

```
java ExamServerGUI
```

You should see:

Exam Server Started...

### Run Client

Open new terminal and run:

```
java ExamClientGUI
```

### Login

Enter:

Username: student1

Password: 1234

Course Name: DBMS

Semester: 5

Click Login & Start Exam

## PHASE 5: Generate JWT Token

Open CMD and run:

```
curl -X POST http://localhost:8080/realms/OnlineExamRealm/protocol/openid-connect/token -H "Content-Type: application/x-www-form-urlencoded" -d "client_id=exam-app&username=student1&password=1234&grant_type=password"
```

Expected Output:

```
"access_token": "eyJhbGciOi..."
```

## PHASE 6: Decode Token

Open:

<https://jwt.io>

Paste access token in left-side Encoded box.

You will see:

- username
- role
- expiry

## Important Viva Questions

- **What is Docker?** Docker runs Keycloak inside a container.
- **What is Keycloak?** Keycloak is an authentication and access management system.
- **What is JWT Token?** JWT token is a secure authentication token.
- **What is Role-based Access?** Access is granted based on user role like student or admin.
- **What does server do?** Server validates token and grants access.

## Final Conclusion

The experiment successfully demonstrated secure authentication using Docker-based Keycloak with JWT token validation and role-based access control.